

The North American flying squirrel and the Draco lizard of Southeast Asia have both evolved a membrane between their limbs that allows them to glide between trees in their habitats. Given that the species do not share a recent common ancestor, which of the following patterns of evolution would explain the development of this anatomical structure in both organisms?

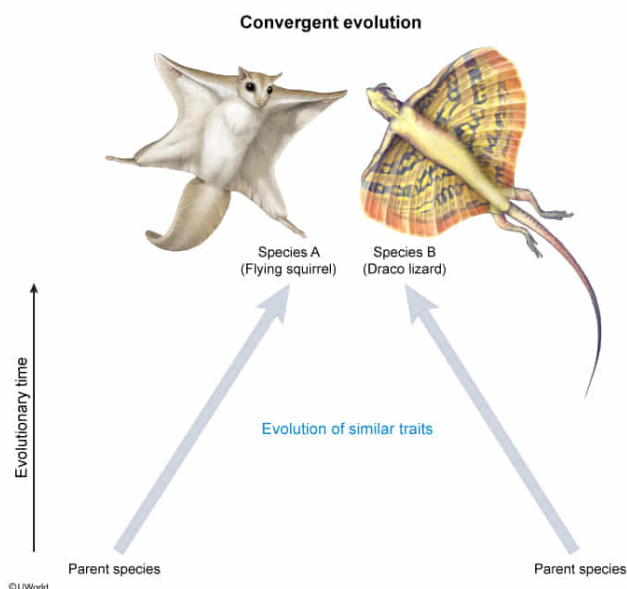
- I. Parallel evolution
- II. Divergent evolution
- III. Convergent evolution

- ☐ A. I only [27%]
- ☐ B. II only [4%]
- ✓ ☒ C. III only [61%]
- ☐ D. I and II only [6%]

Correct

60% answered correctly

Explanation:



According to the theory of **natural selection**, organisms possessing traits that are best suited to their environment are more likely to survive and reproduce (ie, pass on their genes) than those with less favorable traits. As a result, traits/phenotypes that increase chances of survival prevail in the population, causing the species to adapt to its environment over time (ie, evolve).

However, the patterns of evolution across time often vary depending on the environmental pressures faced by different species. If two species inhabit

similar environments, they often **evolve analogous (comparable) characteristics** in response to facing **comparable environmental pressures** (eg, habitat, food availability, predation). In contrast, if two species inhabit *distinct* environments with *different* environmental pressures, they are likely to evolve *unique* phenotypes that are best suited to their different environments. Accordingly, the flying squirrel and Draco lizard do not share a recent common ancestor, yet both have evolved an anatomical structure for gliding between trees. Because they both live in tree-containing habitats, it is likely that both species *independently* acquired a membrane between their limbs to adapt to similar environmental pressures and improve their chances of survival. For example, enhanced mobility or "gliding" between trees may have simplified foraging for food or aided in predator avoidance. Because the evolution of this feature occurred independently in these two very **distantly related species**, this is an example of **convergent evolution (Number III)**.

(Number I) **Parallel evolution** occurs when two *more closely* related species (ie, descended from a recent common ancestor) continue to evolve the same characteristics to adapt to *similar* environments. Because the flying squirrel and Draco lizard do *not* share a recent common ancestor, they are *not* displaying parallel evolution.

(Number II) **Divergent evolution** occurs when two species descended from a recent common ancestor inhabit *contrasting* environments and evolve *distinct* characteristics that allow them to better adapt to their differing environmental pressures. The flying squirrel and Draco lizard share similar (*not* distinct) characteristics; therefore, this is *not* an example of divergent evolution.

Educational objective:

Convergent evolution leads to similar characteristics in distantly related species that are exposed to similar environmental pressures. Parallel evolution also leads to similar characteristics in species found in similar environments but occurs in species with a more recent common ancestor. Divergent evolution leads to unique characteristics in somewhat closely related species that face contrasting environmental pressures.

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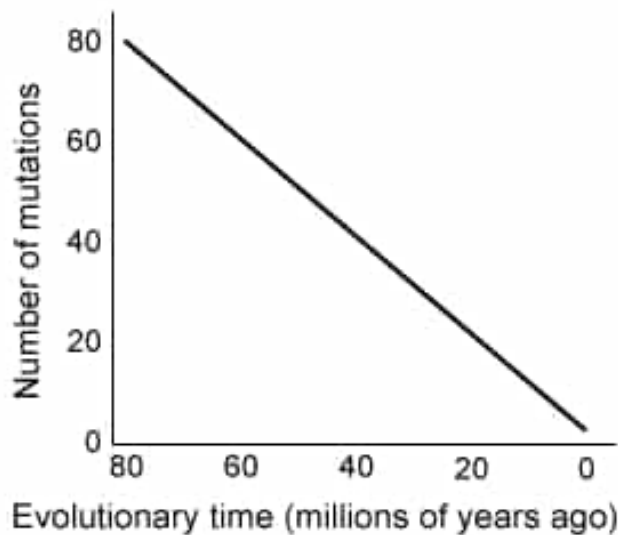
Subject:
Biology

QID:402157

System:
Genetics and Evolution

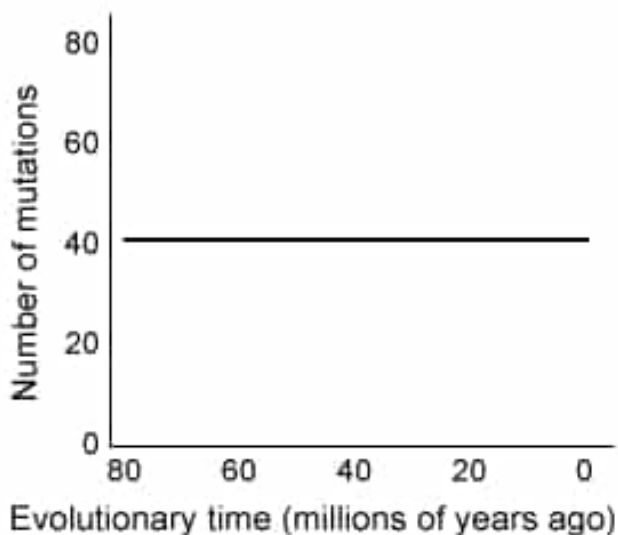
Given that a species diverged from its ancestor 80 million years ago, which of the following graphs most likely illustrates the pattern of mutation accumulation in the genome of this species over time?

☐ A.



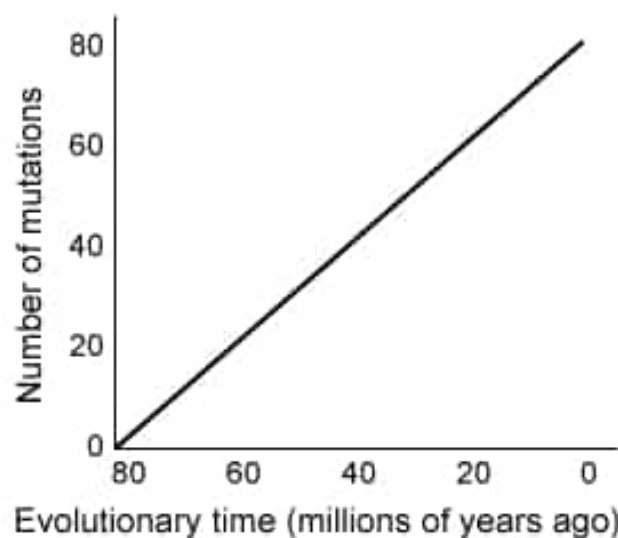
[9%]

☐ B.



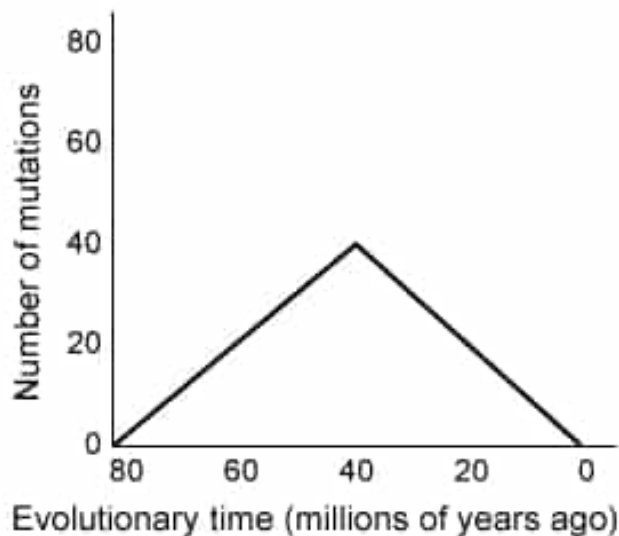
[4%]

✓ ☒ C.



[81%]

☐ D.

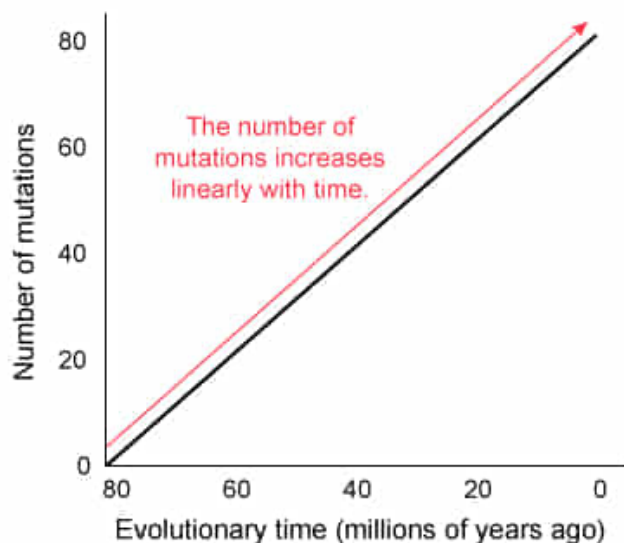


[4%]

Correct

80% answered correctly

Explanation:



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According to the neutral theory of molecular evolution, most genetic mutations are neutral and do not affect the **fitness** (reproductive success) of an organism. **Genetic drift** leads to the random fixation or loss of neutral mutations; ultimately, however, these **neutral mutations accumulate** at a **fairly constant rate** in a species over evolutionary time. This accumulation of neutral mutations in the homologous DNA of species is said to be **linear** and is

referred to as a "molecular clock." The **molecular clock model** allows researchers to measure evolutionary time and estimate the evolutionary relationships between species by analyzing the rate of neutral mutations in the genome.

The species described above diverged from its ancestor 80 million years ago and, according to the molecular clock model, it has likely been accumulating mutations across this entire span of time. Therefore, the overall **number of mutations** should **increase linearly over evolutionary time**; only the graph in **Choice C** shows this linear increase in mutations over time.

(Choices A, B, and D) Because the increase in neutral mutations is linear over time, the species described should have a *higher* (not **lower or unchanged**) number of mutations than it did after diverging from its ancestor.

Educational objective:

Most genetic mutations are neutral and accumulate at a fairly constant rate across organisms over evolutionary time. The overall number of mutations in a species will therefore increase linearly across time. By analyzing the rate of neutral mutations in the genome, this "molecular clock" of evolution can be used to measure evolutionary time and estimate the evolutionary relationships between species.

Time spent:0 sec

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